Author resolution. The homogeneous degree-one affine-line and point classes, the degree-zero unit, and the degree-one generators of the graded ideal are now explicit. This records the existing grading convention and is applied under the author's standing authorization.

Let $!I$ be the ideal of $\mathbf{K}\,\mathrm{RES}$ generated by the elements $[\gamma^\sharp] - [\mathbb{G}_m]$, where $\gamma \in \Gamma$ is definable. The quotient ring $\mathbf{K}\,\mathrm{RES}/!I$ is written as $!\mathbf{K}\,\mathrm{RES}$. In the graded ring, $!I[*]$ is generated by the homogeneous degree-one differences $[\gamma^\sharp] - [\mathbb{G}_m] \in \mathbf{K}\,\mathrm{RES}[1]$, and we set $!\mathbf{K}\,\mathrm{RES}[*] = \mathbf{K}\,\mathrm{RES}[*]/!I[*]$.

The ideal of $\mathbf{K}\,\mu\mathrm{RES}[*]$ generated by the elements

$$[(\mathbb{G}_m, \alpha + \beta)] - [(\alpha^\sharp, \beta)] \in \mathbf{K}\,\mu\mathrm{RES}[1],$$

where $\alpha, \beta \in \Gamma$ are definable, is denoted by $!\mu I[*]$, and the (graded) quotient ring $\mathbf{K}\,\mu\mathrm{RES}[*]/!\mu I[*]$ by $!\mathbf{K}\,\mu\mathrm{RES}[*]$.

The quotient maps from "$\mathbf{K}$" to "$!\mathbf{K}$" will all be denoted by ι, and will often be omitted when it is clear in the context that the latter one is meant. We will also use the same notation for elements when passing from the former to the latter.

Note that the category $\mu\mathrm{RES}[*]$ and the ring $!\mathbf{K}\,\mu\mathrm{RES}[*]$ will be dismissed after Proposition 4.7 as inadequate for our purpose, and a new construction is introduced thereafter.

4.1. **Without volume forms.** The discussion here more or less follows [9, § 2.5] (also see [8, Theorem 10.5]).

For $0 \leq l \leq n$, consider the maps

$$\epsilon_g^{n,l} : \mathbf{K}\,\mathrm{RES}[n-l] \otimes_{\mathbb{Z}} \mathbf{K}\,\Gamma[l] \longrightarrow !\mathbf{K}\,\mathrm{RES}[n]$$

given by

$$x \otimes y \longmapsto \chi_g(y) x [\mathbb{G}_m]^l. \tag{4.1}$$

The component maps are first defined on tensor products over $\mathbb{Z}$. Their sum respects the homogeneous balancing relations over $\mathbf{K}\,\Gamma^{\mathrm{fin}}[*]$ after passing to $!\mathbf{K}\,\mathrm{RES}[*]$: a finite definable Gamma set is a disjoint union of its definable singleton values, and moving such a singleton between the two factors changes its contribution from $[\gamma^\sharp]$ to $[\mathbb{G}_m]$, whose difference vanishes in the target. The resulting tensor assignment also identifies the coefficient-translation pairs of Proposition 3.18. Indeed, if a RES coefficient R admits a definable map $R \longrightarrow \delta^\sharp$, the corresponding pair translates a single coordinate of the Γ factor by δ. Both χ_g and χ_b are invariant under that definable translation. Thus the degree-n sum descends through the kernel congruence of the surjection Ψ, giving a homomorphism

Resolved referee R058 [Clarification] – Tensor products of homogeneous pieces.
Original source line 1442. Strictly speaking, an individual homogeneous piece such as $\mathbf{K}\,\mathrm{RES}[n-l]$ is not a module over the whole graded ring $\mathbf{K}\,\Gamma^{\mathrm{fin}}[*]$. Define the displayed component maps first on $\mathbf{K}\,\mathrm{RES}[n-l] \otimes_{\mathbb{Z}} \mathbf{K}\,\Gamma[l]$, then show that their sum descends through the homogeneous balancing relations in the full graded tensor product. The quotient $!I[*]$ gives exactly the required compatibility.

Author resolution. The component maps are now explicitly over the integers; the subsequent sum is shown to respect the full graded balancing relations. This is a clarification of the construction, applied under the standing instruction for clear corrections.

$$\mathbf{K}\,\mathrm{RV}[n] \longrightarrow !\mathbf{K}\,\mathrm{RES}[n].$$

For later use, denote the resulting graded homomorphisms, before degree normalization, by

$$\widetilde{\mathbb{E}}_g, \widetilde{\mathbb{E}}_b : \mathbf{K}\,\mathrm{RV}[*] \longrightarrow !\mathbf{K}\,\mathrm{RES}[*], \qquad \widetilde{\mathbb{E}}_\chi\big(\Psi(x \otimes y)\big) = \chi(y) x [\mathbb{G}_m]^l \quad (y \in \mathbf{K}\,\Gamma[l]).$$

No further quotient of the RES target is required for this descent. Composing this with the homomorphism

$$\epsilon_g^{n,m} : !\mathbf{K}\,\mathrm{RES}[n] \longrightarrow !\mathbf{K}\,\mathrm{RES}[m] : x \longmapsto x[\mathbb{A}]^{m-n}$$

and taking the direct sum of the resulting homomorphisms over all n with $n \leq m$, we obtain a homomorphism

$$\epsilon_g^m : \mathbf{K}\,\mathrm{RV}[\leq m] \longrightarrow !\mathbf{K}\,\mathrm{RES}[m].$$

These homomorphisms ϵ_g^m, $m = 0, 1, \ldots$, are compatible in the obvious sense. Consequently, they give rise to a homomorphism

$$\mathbb{E}_g : \mathbf{K}\,\mathrm{RV}[*] \longrightarrow !\mathbf{K}\,\mathrm{RES}[*][[\mathbb{A}]^{-1}]$$

whose range is precisely the zeroth piece $(!\mathbf{K}\,\mathrm{RES}[*][[\mathbb{A}]^{-1}])_0$ of the $\mathbb{Z}$-graded ring on the righthand side. We have

$$\mathbb{E}_g([\mathrm{RV}^{\circ\circ}] + 1 - [1]) = -[\mathbb{G}_m][\mathbb{A}]^{-1} + 1 - [1][\mathbb{A}]^{-1} = 0. \tag{4.2}$$

All terms have degree zero: the degree-one point maps to $[1][\mathbb{A}]^{-1}$. The cancellation uses the homogeneous identity $[\mathbb{A}] = [\mathbb{G}_m] + [1]$.

Resolved referee R059 [Error] – Missing degree-one point factor.
Original source line 1461. In the graded target, $\mathbb{E}_g([1]) = [1][\mathbb{A}]^{-1}$, not $[\mathbb{A}]^{-1}$. Thus the correct computation is $-[\mathbb{G}_m][\mathbb{A}]^{-1} + 1 - [1][\mathbb{A}]^{-1} = 0$, using the homogeneous identity $[\mathbb{A}] = [\mathbb{G}_m] + [1]$. As written, the final term has degree -1, while the first two have degree 0. The formula $\mathbb{E}_g([1]) = [\mathbb{A}]^{-1}$ becomes correct only after forgetting the grading.

Author resolution. The missing degree-one point factor is restored. The corrected cancellation follows from the homogeneous affine-line identity; its algebraic calculation is checked in Lean.

Thus $\mathbb{E}_g$ induces an eponymous homomorphism

$$\mathbb{E}_g : \mathbf{K}\,\mathrm{RV}[*]/(\boldsymbol{P} - 1) \longrightarrow !\mathbf{K}\,\mathrm{RES}[*][[\mathbb{A}]^{-1}].$$

There is a parallel construction, where we replace χ_g with χ_b and $[\mathbb{A}]^{-1}$ with $[1]^{-1}$. The latter replacement is needed so to make $(\boldsymbol{P} - 1)$ vanish as in (4.2). The resulting homomorphism is

$$\mathbb{E}_b : \mathbf{K}\,\mathrm{RV}[*]/(\boldsymbol{P} - 1) \longrightarrow !\mathbf{K}\,\mathrm{RES}[*][[1]^{-1}], \tag{4.3}$$

whose range is precisely the zeroth piece $(!\mathbf{K}\,\mathrm{RES}[*][[1]^{-1}])_0$.

Remark 4.4. The zeroth graded piece $(!\mathbf{K}\,\mathrm{RES}[*][[\mathbb{A}]^{-1}])_0$ of $!\mathbf{K}\,\mathrm{RES}[*][[\mathbb{A}]^{-1}]$ is canonically isomorphic to the colimit of the groups $!\mathbf{K}\,\mathrm{RES}[n]$ with the transition morphisms $\epsilon_g^{n,m}$, which is actually what appears in the construction above. So there is a natural homomorphism from $(!\mathbf{K}\,\mathrm{RES}[*][[\mathbb{A}]^{-1}])_0$ to the similar-looking ring $!\mathbf{K}\,\mathrm{RES}[[\mathbb{A}]^{-1}]$, induced by the assignment $x \longmapsto x[\mathbb{A}]^{-n}$ for $x \in !\mathbf{K}\,\mathrm{RES}[n]$. Note that this homomorphism is not surjective.

Resolved referee R060 [Error; revised] – The asserted injection is false.
Original source line 1474. The natural homomorphism in this sentence need not be injective, already over an untwisted residue field $\mathbb{C}$. To distinguish the two constructions precisely, let R_n be the Grothendieck group of complex constructible sets of dimension at most n, with scissor relations imposed *within that category*, and set $R = \bigoplus_{n\geq 0} R_n$. This is the graded RES ring in the untwisted case; R_n is not by definition the nth dimension-filtration subgroup of the ordinary Grothendieck ring. Let $\mathbb{L}_1 = [\mathbb{A}^1]_1$. Ito–Miura–Okawa–Ueda give smooth projective threefolds X, Y that are not stably birational but satisfy $([X] - [Y])\mathbb{L} = 0$ in the ordinary ring (Theorems 1.1 and 1.3). Thus $z = ([X]_3 - [Y]_3)\mathbb{L}_1^{-3}$ maps to zero in $K_0(\mathrm{Var}_{\mathbb{C}})[\mathbb{L}^{-1}]$. Yet $z \neq 0$: for every $m \geq 0$, the additive invariant on R_{3+m} recording its top-dimensional birational classes distinguishes $X \times \mathbb{A}^m$ and $Y \times \mathbb{A}^m$. Hence no homogeneous power of $\mathbb{L}_1$ kills the numerator. To fit the paper's parameter conventions, take the full algebraically closed Puiseux

field over $\mathbb{C}$ as the parameter field. Every definable value then has a named RV lift, so all $!I$ generators already vanish and $\mathrm{RES}[n]$ identifies with the indicated constructible-set category. Replace "injective homomorphism" by "natural homomorphism." The comparison of $\mathbb{E}_g$ with $\mathbb{E}_b$ uses only that natural map, and the point-localized identification in the next paragraph is unaffected.

Author resolution. The construction supplies the natural comparison map, and no later argument here uses its injectivity. The independent top-dimensional obstruction is recorded in the section review; the point-localized isomorphism is unchanged.

Similarly, there is an injective homomorphism $(!\mathbf{K}\,\mathrm{RES}[*][[1]^{-1}])_0 \longrightarrow !\mathbf{K}\,\mathrm{RES}[[1]^{-1}] \cong !\mathbf{K}\,\mathrm{RES}$, and this is indeed an isomorphism.

If the codomain of $\mathbb{E}_b$ is changed to $!\mathbf{K}\,\mathrm{RES}[[\mathbb{A}]^{-1}]$ in the obvious way then it may be compared with $\mathbb{E}_g$. They are different since

$$\mathbb{E}_b([1]) = 1 \quad \text{and} \quad \mathbb{E}_g([1]) = \mathbb{E}_g([\mathrm{RV}^{\circ\circ}]) + 1 = -[\mathbb{G}_m][\mathbb{A}]^{-1} + 1 = [\mathbb{A}]^{-1}. \tag{4.4}$$

We can equalize them by forcing $[\mathbb{A}] = 1$ (hence $\mathbb{E}_g(x \otimes y) = \mathbb{E}_b(x \otimes y) = 0$ for homogeneous $y \in \mathbf{K}\,\Gamma[l]$ with $l > 0$). The (complex version of the) resulting homomorphism is indeed the one constructed in [9, (2.5.7)].

The homomorphism $\mathbb{E}_b$ will be used in constructing the motivic Milnor fiber below, but not $\mathbb{E}_g$. It is not that $\mathbb{E}_g$ would not work; it is just that, conceptually, $\mathbb{E}_b$ is a better choice, see Remarks 4.16 and 8.14.

Resolved referee R168 [Typo; new] – Missing article.
Original source line 1485. Write "used in constructing *the* motivic Milnor fiber below."

Author resolution. The missing article is inserted under the author's standing instruction for clear corrections.

4.2. **With volume forms.** Let $\mathrm{FN}(\Gamma, \mathbb{Z})$ be the set of functions $f : \Gamma \longrightarrow \mathbb{Z}$ such that the range of f is finite and $f^{-1}(m)$ is definable for every $m \in \mathbb{Z}$. Addition in $\mathrm{FN}(\Gamma, \mathbb{Z})$ is defined in the obvious way. Multiplication in $\mathrm{FN}(\Gamma, \mathbb{Z})$ is given by the convolution product as follows. For $f, g \in \mathrm{FN}(\Gamma, \mathbb{Z})$ and $\gamma \in \Gamma$, let h_γ be the function $\Gamma \longrightarrow \mathbb{Z}$ given by $\alpha \longmapsto f(\alpha)g(\gamma - \alpha)$. The range of h_γ is obviously finite. For each $m \in \mathbb{Z}$, $h_\gamma^{-1}(m)$ is a finite disjoint union of sets of the form

$$f^{-1}(m') \cap (\gamma - g^{-1}(m'')),$$

where $m'm'' = m$, and hence is γ-definable. Let

$$h_\gamma^* = \sum_m m\chi(h_\gamma^{-1}(m)) \in \mathbb{Z}$$

and $f * g : \Gamma \longrightarrow \mathbb{Z}$ be the function given by $\gamma \longmapsto h_\gamma^*$. It is not hard to see that, by o-minimality in the Γ-sort, $f * g = g * f$ and it belongs to $\mathrm{FN}(\Gamma, \mathbb{Z})$. Thus $\mathrm{FN}(\Gamma, \mathbb{Z})$ is a commutative ring. Its multiplicative identity is p_0. Associativity is checked on the additive generators in Notation 4.5. Writing $rq_\beta = \eta r$, the mixed r, q, q check reduces to $\eta^2 = -\eta$, which holds for both $\eta = -1$ and $\eta = 0$; the other generator triples follow directly from the displayed rules.

Resolved referee R061 [Clarification; revised] – Associativity of convolution.
Original source line 1501. Optional proof detail: add that p_0 is the multiplicative identity and that associativity follows by checking the additive generators in Notation 4.5. For example, writing $rq_\beta = \eta r$, the only nontrivial r, q, q check reduces to $\eta^2 = -\eta$, which holds for both $\eta = -1$ and $\eta = 0$. The ring assertion is correct for both choices of χ.

Author resolution. The identity and generator associativity check are now explicit. This supplies an omitted justification without changing either convolution product.

Notation 4.5. In $\mathrm{FN}(\Gamma,\mathbb{Z})$, let r be the constant function 1 and, for each definable element $\gamma\in\Gamma$, let p_γ, q_γ be the characteristic functions of γ, (γ,∞), respectively. By o-minimality, $\mathrm{FN}(\Gamma,\mathbb{Z})$ is generated as a $\mathbb{Z}$-module by elements of the forms r, p_γ, q_γ. We have the following equalities:

$$rp_\beta = r,\quad p_\alpha p_\beta = p_{\alpha+\beta},\quad p_\alpha q_\beta = q_{\alpha+\beta},\quad q_\alpha q_\beta = -q_{\alpha+\beta}.$$

In addition,

- if $\chi=\chi_g$ then $r^2 = rq_\beta = -r$,
- if $\chi=\chi_b$ then $r^2 = r$ and $rq_\beta = 0$.

Remark 4.6. For each $\boldsymbol{I}=(I,\mu)\in\mu\Gamma[k]$, let $\mu_I : I\longrightarrow\Gamma$ be the function given by $\gamma\longmapsto\mu(\gamma)+\Sigma\gamma$ and $\lambda_{\boldsymbol{I}}:\Gamma\longrightarrow\mathbb{Z}$ the function given by $\gamma\longmapsto\chi(\mu_I^{-1}(\gamma))$. By o-minimality, the range of $\lambda_{\boldsymbol{I}}$ is finite and, for every $m\in\mathbb{Z}$, $\lambda_{\boldsymbol{I}}^{-1}(m)$ is definable. So $\lambda_{\boldsymbol{I}}\in\mathrm{FN}(\Gamma,\mathbb{Z})$. If $F:(I,\mu)\longrightarrow(J,\sigma)$ is a $\mu\Gamma[k]$-morphism then F restricts to a bijection from $\mu_I^{-1}(\gamma)$ to $\sigma_J^{-1}(\gamma)$ for every $\gamma\in\Gamma$ and hence $\lambda_{\boldsymbol{I}}=\lambda_{\boldsymbol{J}}$ and we may write $\lambda_{\boldsymbol{I}}$ as $\lambda_{[\boldsymbol{I}]}$. Let $\boldsymbol{J}=(J,\sigma)\in\mu\Gamma[l]$. If $k=l$ then clearly $\lambda_{[\boldsymbol{I}]}+\lambda_{[\boldsymbol{J}]}=\lambda_{[\boldsymbol{I}]+[\boldsymbol{J}]}$. An easy computation shows that, for every $\gamma\in\Gamma$,

$$(\lambda_{[\boldsymbol{I}]}*\lambda_{[\boldsymbol{J}]})(\gamma)=\chi((\mu\sigma)^{-1}_{I\times J}(\gamma)),$$

where $\mu\sigma: I\times J\longrightarrow\Gamma$ is the function given by $(\alpha,\beta)\longmapsto\mu(\alpha)+\sigma(\beta)$, and hence $\lambda_{[\boldsymbol{I}]}*\lambda_{[\boldsymbol{J}]}=\lambda_{[\boldsymbol{I}][\boldsymbol{J}]}$. Put $\Gamma_{\mathrm{def}}=\Gamma\cap\mathrm{dcl}^{\mathrm{eq}}(\mathbb{S})$ and let $\mathbb{Z}[\Gamma_{\mathrm{def}}]$ be its group ring over $\mathbb{Z}$, viewed as the subring of $\mathrm{FN}(\Gamma,\mathbb{Z})$ spanned by p_γ for all $\gamma\in\Gamma_{\mathrm{def}}$. The degree-zero objects have definable constant weights, so $\mathbf{K}\,\mu\Gamma[0]\cong\mathbb{Z}[\Gamma_{\mathrm{def}}]$. It follows that the map

Resolved referee R062 [Error] – The group ring uses definable value-group constants.
Original source line 1520. The degree-zero coefficient group is the group of *definable* elements of Γ, not necessarily $\Gamma(\mathbb{S})$. With the present conventions, put $\Gamma_{\mathrm{def}}=\Gamma\cap\mathrm{dcl}^{\mathrm{eq}}(\mathbb{S})$ and replace $\mathbb{Z}\Gamma$ throughout this construction by $\mathbb{Z}[\Gamma_{\mathrm{def}}]$; in the usual valued-field-generated setting $\Gamma_{\mathrm{def}}=\mathbb{Q}\otimes\Gamma(\mathbb{S})$. For example, over $\mathbb{C}((t))$, the singleton object of $\mu\Gamma[0]$ with constant volume form $\mathrm{val}(t)/2$ is allowed and maps to $p_{1/2}$, although $\Gamma(\mathbb{S})=\mathbb{Z}$. Thus the stated target misses legitimate degree-zero classes. This distinction is already acknowledged in the remark following Definition 3.13.

Author resolution. Degree-zero objects permit all definable constant weights. The coefficient group is now consistently the group of definable Gamma elements, as required by the earlier Gamma-category convention.

$$\lambda:\mathbf{K}\,\mu\Gamma[*]\longrightarrow\mathbb{Z}[\Gamma_{\mathrm{def}}]\oplus X\,\mathrm{FN}(\Gamma,\mathbb{Z})[X]$$

determined by the assignment

$$[\boldsymbol{I}]\longmapsto\lambda_{[\boldsymbol{I}]}X^k,\quad \boldsymbol{I}\in\mu\Gamma[k],$$

is a graded ring homomorphism. Of course, there are two such homomorphisms λ_g and λ_b, corresponding to the two choices $\chi=\chi_g$ and $\chi=\chi_b$.

We claim that the assignments for $k\geq 1$

$$p_\gamma X^k\longmapsto[\mathbb{G}_m]^{k-1}[(\mathbb{G}_m,\gamma)],\quad q_\gamma X^k\longmapsto[\mathbb{G}_m]^{k-1}[(\{1\},\gamma)],\quad rX^k\longmapsto 0 \tag{4.5}$$

induce a graded ring homomorphism

$$\psi:\mathbb{Z}[\Gamma_{\mathrm{def}}]\oplus X\,\mathrm{FN}(\Gamma,\mathbb{Z})[X]\longrightarrow{}!\mathbf{K}\,\mu\mathrm{RES}[*]/([\mathbb{A}]).$$

Note that $\mathbb{Z}[\Gamma_{\mathrm{def}}]$ may be regarded as a subring of $\mathbf{K}\,\mu\mathrm{RES}[0]$ by identifying p_γ with $[(\infty,\gamma)]$, which is the intended interpretation of the first assignment for $k=0$. The $q_\gamma X^k$ and rX^k assignments apply only for $k\geq 1$. For the claim, it suffices to check that ψ obeys the equational constraints for the generators above in Notation 4.5, which are all straightforward (the equality $\psi(q_\alpha X^k)\psi(q_\beta X^l)=\psi(-q_{\alpha+\beta}X^{k+l})\quad(k,l\geq 1)$ holds since $[\mathbb{A}]=[\mathbb{G}_m]+[1]=0$ is forced).

Resolved referee R063 [Typo] – Restore the degrees in the multiplicativity check.
Original source line 1544. The symbols q_α alone are not degree-zero elements of the displayed domain. State the check as $\psi(q_\alpha X^k)\psi(q_\beta X^l) = \psi(-q_{\alpha+\beta}X^{k+l})$ for $k, l \geq 1$. Also state explicitly that the second and third assignments in (4.5) apply only for $k \geq 1$; the separate interpretation at $k = 0$ applies to p_γ.

Author resolution. The positive-degree domain of the ray and constant assignments and the degrees in the multiplicativity check are now explicit.

Let $\mathfrak{h}$ denote the element $[(\mathrm{RV}^{\circ\circ}, -\mathrm{vrv})] \in \mathbf{K}\,\mu\mathrm{RV}[1]$.

Proposition 4.7. *There are two graded ring homomorphisms*

$$\mu\mathbb{E}_g, \mu\mathbb{E}_b : \mathbf{K}\,\mu\mathrm{RV}[*] \longrightarrow {!}\mathbf{K}\,\mu\mathrm{RES}[*]/([\mathbb{A}])$$

such that

- $\boldsymbol{P} \in \mathbf{K}\,\mu\mathrm{RV}[1]$ *vanishes under both of them,*
- *for all* $x \in \mathbf{K}\,\mu\mathrm{RES}[*]$, $\mu\mathbb{E}_g(x) = \mu\mathbb{E}_b(x) = x/([\mathbb{A}])$,
- *they are distinguished by* $\mu\mathbb{E}_g(\mathfrak{h}) = +[1]/([\mathbb{A}])$ *and* $\mu\mathbb{E}_b(\mathfrak{h}) = 0$.

Resolved referee R064 [Error] – Sign of the geometric retraction on h.
Original source line 1560. The construction gives the opposite sign in the third bullet. For $\boldsymbol{I} = ((0,\infty), -\mathrm{id})$, the corrected weight μ_I is identically zero, so $\lambda_g([\boldsymbol{I}]) = -p_0X$. Consequently $\mu\mathbb{E}_g(\mathfrak{h}) = \psi(-p_0X) = -[\mathbb{G}_m] = +[1]$ modulo $([\mathbb{A}])$. The bounded value is indeed zero. Replace $-[1]/([\mathbb{A}])$ by $+[1]/([\mathbb{A}])$. This confirms the sign concern in the supplied human referee report; repeating the sign printed in the cited HK theorem does not fix its incompatibility with the present definitions.

Author resolution. The sign is corrected directly from the stated construction: the corrected weight is zero, the geometric ray contributes $-p_0X$, and its image is $-[\mathbb{G}_m] = +[1]$ modulo the affine-line class. The bounded value remains zero.

Proof. The product map

$$(\iota/([\mathbb{A}])) \times (\psi \circ \lambda) : \mathbf{K}\,\mu\mathrm{RES}[*] \times \mathbf{K}\,\mu\Gamma[*] \longrightarrow {!}\mathbf{K}\,\mu\mathrm{RES}[*]/([\mathbb{A}])$$

is $\mathbf{K}\,\mu\Gamma^{\mathrm{fin}}[*]$-bilinear (because of the ideal ${!}\mu I[*]$ in the target). It also identifies the measured coefficient-translation pairs of Proposition 3.18. For a translation $\eta = \gamma + \delta e_i$ of (I, σ), the target volume form is

$$\sigma_\delta(\eta) = \sigma(\eta - \delta e_i) - \delta.$$

Consequently

$$\sigma_\delta(\eta) + \Sigma\eta = \sigma(\gamma) + \Sigma\gamma.$$

The corrected-weight fibers are therefore definably bijective, so their two Euler characteristics agree and λ has the same value on the pair. The balanced product map thus descends through the kernel congruence of the surjection $\mu\Psi$, giving the desired graded homomorphisms. To compute their values, use the explicit tensor preimages

$$\mu\Psi\big([1] \otimes 1 - 1 \otimes [(0,\infty)]\big) = \boldsymbol{P}, \qquad \mu\Psi\big(1 \otimes [((0,\infty), -\mathrm{id})]\big) = \mathfrak{h}.$$

These formulas use preimages under the product map. For the value on $\mathfrak{h}$, the corrected weight on $((0,\infty), -\mathrm{id})$ is zero, so

$$\lambda_g([((0,\infty), -\mathrm{id})]) = -p_0X, \qquad \mu\mathbb{E}_g(\mathfrak{h}) = \psi(-p_0X) = -[\mathbb{G}_m] = +[1] \pmod{([\mathbb{A}])}.$$

For χ_b the open ray has characteristic zero, giving $\mu\mathbb{E}_b(\mathfrak{h}) = 0$.

Finally, [1] is nonzero in the target. After forgetting degree and volume forms, choose auxiliary RV representatives for the finitely many value fibers of each RES object. Normalization and quantifier elimination give pure residue-field definable sets, and hence a homomorphism to the

Grothendieck ring of pure $\Bbbk$-definable sets modulo the affine-line class $\mathbb{L}$. The fiber relations vanish under this homomorphism, $[\mathbb{A}]$ maps to $\mathbb{L}$, and $[1]$ maps to 1. The latter quotient is nonzero: any proposed equality $1 = 0$ uses only finitely many formulas, parameters and definable bijections. By the Lefschetz principle these data specialize to an algebraic closure of a finite field, with all parameters in some $\mathbb{F}_q$. A definable bijective relation over $\mathbb{F}_q$ induces a bijection on its $\mathbb{F}_q$-points. Counting these points modulo q preserves the defining relations, sends $\mathbb{L}$ to 0 and sends 1 to 1, a contradiction. Thus the two displayed values on $\mathfrak{h}$ are distinct.

Resolved referee R065 [Typo] – Agreement.
Original source line 1570. Change the final verb "follow" to "follows," since the subject is "The existence."

Author resolution. The subject–verb agreement is corrected under the author's standing instruction for clear corrections.

□

The discussion above is more or less an expanded version of the proof of [8, Theorem 10.11].

Remark 4.8. Alternatively, for $\chi = \chi_g$, we may replace $rX^k \longmapsto 0$ in (4.5) with $rX^k \longmapsto [\mathbb{G}_m]^{k-1}[1]$, that is, sending rX^k and q_0X^k to the same element. Let J be the enlargement of $!\mu I[*]$ by the elements $[1] - [(\{1\}, \gamma)] \in \mathbf{K}\,\mu\mathrm{RES}[1]$. Then ψ is still a graded ring homomorphism with this enlarged quotient. Consequently, for instance, we can construct a ring homomorphism

$$\mathbf{K}\,\mu\mathrm{RV}[*] \longrightarrow \mathbf{K}\,\mu\mathrm{RES}[*]/(J + ([\mathbb{A}])) \xrightarrow{\phi} !\mathbf{K}\,\mathrm{RES}[*]/([\mathbb{A}]), \tag{4.6}$$

where the homomorphism ϕ is induced by the obvious forgetful functor.

Resolved referee R066 [Clarification; revised] – Name the enlarged quotient.
Original source line 1580. Optional notation clarification: the preceding sentence already stipulates enlargement of $!\mu I[*]$. In the alternative construction, name the enlarged ideal J and write the middle term of (4.6) as $\mathbf{K}\,\mu\mathrm{RES}[*]/(J + ([\mathbb{A}]))$, so that the old and enlarged quotients are visibly distinguished. Read with that stipulated enlargement, the construction is consistent. The original assignment $rX^k \mapsto 0$ needs no such additional relations.

Author resolution. The already stipulated enlarged ideal is named J and the alternative target displays J explicitly; no additional relation is imposed on the original construction.

We shall not, however, pursue this further, since taking the quotient by the ideal $([\mathbb{A}])$ is detrimental to our purpose here. To avoid it, we need to work with the categories of doubly bounded objects.

Definition 4.9 (Doubly bounded RV-categories)**.** An object $(U, f) \in \mathrm{RV}[k]$ is *doubly bounded* if U is. Denote the full subcategory of $\mathrm{RV}[k]$ of doubly bounded objects by $\mathrm{RV}^{\mathrm{db}}[k]$, and similarly for the categories $\mu\mathrm{RV}^{\mathrm{db}}[k]$, $\mu\Gamma^{\mathrm{db}}[k]$.

Remark 4.10. By Lemma 2.41, if $(U, f, \omega) \in \mu\mathrm{RV}^{\mathrm{db}}[k]$ then $f(U)$, $\omega(U)$ are both doubly bounded as well. The product construction gives a homomorphism

$$\mu\Psi^{\mathrm{db}} : \mathbf{K}_+\,\mu\mathrm{RES}[*] \otimes \mathbf{K}_+\,\mu\Gamma^{\mathrm{db}}[*] \longrightarrow \mathbf{K}_+\,\mu\mathrm{RV}^{\mathrm{db}}[*].$$

We use it below to compute images of explicit tensor classes. The bounded retraction itself is obtained by restricting the already constructed graded map $\widetilde{\mathbb{E}}_b$ after forgetting volume forms.

Notation 4.11. Let $\mathrm{FN}^{\mathrm{db}}(\Gamma, \mathbb{Z})$ be the subgroup of $\mathrm{FN}(\Gamma, \mathbb{Z})$ consisting of functions whose nonzero locus is doubly bounded. For each definable $\gamma \in \Gamma$, set

$$o_\gamma = \begin{cases} \mathbf{1}_{(0,\gamma)}, & \gamma > 0, \\ \mathbf{1}_{(\gamma,0)}, & \gamma < 0, \\ -p_0, & \gamma = 0. \end{cases}$$

As a $\mathbb{Z}$-module, $\mathrm{FN}^{\mathrm{db}}(\Gamma, \mathbb{Z})$ is generated by p_γ, o_γ. For $\gamma > 0$ we have $o_\gamma = q_0 - q_\gamma - p_\gamma$, and for $\gamma < 0$ we have $o_\gamma = q_\gamma - q_0 - p_0$. The convention $o_0 = -p_0$ is a convention for functions; it does not identify o_0 with the characteristic function of the empty interval $(0, 0)$. The convolution products are given by the following cases.

Resolved referee R067 [Error] – Negative interval generators and bounded support.
Original source line 1600. For $\gamma < 0$, the interval $(0, \gamma)$ is empty, so the definition as written cannot yield the claimed generators or quotient: the relation $o_\gamma + p_0$ would even kill the unit p_0. Define $o_\gamma = \mathbf{1}_{(\gamma,0)}$ when $\gamma < 0$, retaining $o_\gamma = \mathbf{1}_{(0,\gamma)}$ when $\gamma > 0$ and $o_0 = -p_0$. Also define $\mathrm{FN}^{\mathrm{db}}(\Gamma, \mathbb{Z})$ explicitly as the subgroup of functions whose nonzero locus is doubly bounded. These conventions make the subsequent formulas meaningful.

Author resolution. The bounded nonzero-support convention and both signs of the interval are now explicit, including the separate function convention at zero. The literal negative interval cannot supply the intended generator.

For $\beta \neq 0$, put $a = \min(\alpha, \alpha + \beta)$ and $b = \max(\alpha, \alpha + \beta)$. Then $p_\alpha o_\beta$ is the characteristic function of (a, b), and

$$p_\alpha o_\beta = \begin{cases} o_b - o_a - p_a, & 0 \leq a < b, \\ o_a - o_b - p_b, & a < b \leq 0, \\ o_a + o_b + p_0, & a < 0 < b. \end{cases}$$

For $\beta = 0$, $p_\alpha o_0 = -p_\alpha$. Also,

$$o_\alpha o_\beta = \begin{cases} -o_{\alpha+\beta}, & \alpha, \beta \geq 0 \text{ or } \alpha, \beta \leq 0, \\ -o_\alpha - o_\beta - p_0, & \alpha < 0 < \beta \text{ or } \beta < 0 < \alpha. \end{cases}$$

Resolved referee R068 [Clarification] – Specify which interval identity applies.
Original source line 1605. After correcting the negative-interval convention, the alternatives can be made unambiguous as follows. Set $a = \min(\alpha, \alpha + \beta)$ and $b = \max(\alpha, \alpha + \beta)$. For $\beta \neq 0$, $p_\alpha o_\beta$ is the characteristic function of (a, b): it is $o_b - o_a - p_a$ if $0 \leq a < b$, $o_a - o_b - p_b$ if $a < b \leq 0$, and $o_a + o_b + p_0$ if $a < 0 < b$. For $\beta = 0$ it equals $-p_\alpha$. Likewise $o_\alpha o_\beta = -o_{\alpha+\beta}$ when $\alpha, \beta \geq 0$ or $\alpha, \beta \leq 0$, and equals $-o_\alpha - o_\beta - p_0$ when α and β have opposite strict signs.

Author resolution. The original alternatives are now assigned their sign and endpoint cases, with the zero interval parameter treated separately.

Notation 4.12. The homomorphism

$$\lambda : \mathbf{K}\,\mu\Gamma^{\mathrm{db}}[*] \longrightarrow \mathbb{Z}[\Gamma_{\mathrm{def}}] \oplus X\,\mathrm{FN}^{\mathrm{db}}(\Gamma, \mathbb{Z})[X]$$

is constructed as before (there is only one such homomorphism now since χ_g, χ_b agree on doubly bounded sets).

Notation 4.13. Recall that $\mathrm{RV}^{\circ\circ}_\gamma = \mathrm{rv}(\mathcal{M}_\gamma \smallsetminus 0)$ (Notation 2.5). Let D_{RV} consist of the nonnegative definable γ such that $\gamma^\sharp$ contains an $\mathbb{S}$-definable point. For each $\gamma \in D_{\mathrm{RV}}$, let

$$\boldsymbol{P}_\gamma = [\mathrm{RV}^{\circ\circ} \smallsetminus \mathrm{RV}^{\circ\circ}_\gamma] + [\{t_\gamma\}] - [1] \in \mathbf{K}\,\mathrm{RV}^{\mathrm{db}}[1],$$

where $t_\gamma \in \gamma^\sharp$ is any definable point. It also stands for the corresponding element in $\mathbf{K}\,\mu\mathrm{RV}^{\mathrm{db}}[1]$ (with the constant volume form 0). Note that, although $[\{t_\gamma\}] = [1]$ in $\mathbf{K}\,\mathrm{RV}^{\mathrm{db}}[1]$, $[\{t_\gamma\}] \neq [1]$ in $\mathbf{K}\,\mu\mathrm{RV}^{\mathrm{db}}[1]$ unless $\gamma = 0$.

Resolved referee R069 [Error] – The radius must admit a definable RV lift.
Original source line 1621. Not every definable γ admits the definable point t_γ required here. Restrict the definition to nonnegative γ for which $\gamma^\sharp$ contains an $\mathbb{S}$-definable point, equivalently $\gamma \in \mathrm{vrv}(\mathrm{RV}(\mathrm{dcl}\,\mathbb{S}))$. In the later valued-field-generated setting this is the value group of the parameter field, rather than its divisible hull. For example, over $\mathbb{C}((t))$, $\gamma = 1/2$ is definable but there is no definable RV element of that value. Make the same restriction in the generating set of $(\boldsymbol{P}_\Gamma)$. The later specialization to integral radii should refer back to this restriction.

Author resolution. The author approved the restriction to nonnegative radii admitting an RV lift definable over the original parameter base. The set D_{RV} now indexes both the generators and the generated ideal.

Clearly $\boldsymbol{P}_\gamma$ does not depend on the choice of $t_\gamma \in \gamma^\sharp$. The ideal of $\mathbf{K}\,\mu\mathrm{RV}^{\mathrm{db}}[*]$ generated by the elements $\boldsymbol{P}_\gamma$ with $\gamma \in D_{\mathrm{RV}}$ is denoted by $(\boldsymbol{P}_\Gamma)$. A definable Gamma value need not have a definable RV lift, as already noted after Definition 3.13. All choices are made over the original parameter base. The images of $(\boldsymbol{P}_\Gamma)$ under the obvious forgetful homomorphisms

$$\mathbf{K}\,\mu\mathrm{RV}^{\mathrm{db}}[*] \longrightarrow \mathbf{K}\,\mathrm{RV}[*], \quad \mathbf{K}\,\mu\mathrm{RV}^{\mathrm{db}}[*] \longrightarrow \mathbf{K}\,\mu\mathrm{RV}[*]$$

are contained in $(\boldsymbol{P} - 1)$, $(\boldsymbol{P})$, respectively.

Resolved referee R070 [Typo] – Closing punctuation.
Original source line 1627. End this sentence with a full stop, not a comma.

Author resolution. The sentence now ends with a full stop.

Remark 4.14. Let $!P$ denote the ideal of $\mathrm{FN}^{\mathrm{db}}(\Gamma, \mathbb{Z})$ generated by the elements $p_\gamma - p_0$, $o_\gamma + p_0$. By the computation in Notation 4.11, $\mathrm{FN}^{\mathrm{db}}(\Gamma, \mathbb{Z})/!P \cong \mathbb{Z}$; the quotient homomorphism is denoted by χ because a simple computation shows that if $\boldsymbol{I} = (I, \mu) \in \mu\Gamma^{\mathrm{db}}[*]$ then indeed $\lambda_{\boldsymbol{I}}/!P = \chi(I)$. Now it is straightforward to check that the assignments

$$p_\gamma X^k \longmapsto [\mathbb{G}_m]^k, \quad o_\gamma X^k \longmapsto -[\mathbb{G}_m]^k \tag{4.7}$$

induce a graded ring homomorphism

$$\psi^{\mathrm{db}} : \mathbb{Z}[\Gamma_{\mathrm{def}}] \oplus X\,\mathrm{FN}^{\mathrm{db}}(\Gamma, \mathbb{Z})[X] \longrightarrow !\mathbf{K}\,\mathrm{RES}[*].$$

In a nutshell, the homomorphism $\psi^{\mathrm{db}} \circ \lambda$ is a "forgetful" map given by $[(I, \mu)] \longmapsto \chi(I)[\mathbb{G}_m]^k$, $(I, \mu) \in \mu\Gamma^{\mathrm{db}}[k]$.

The forgetful homomorphism $\phi : \mathbf{K}\,\mu\mathrm{RES}[*] \longrightarrow !\mathbf{K}\,\mathrm{RES}[*]$ is similar to the one in (4.6).

Proposition 4.15. *There is a graded ring homomorphism*

$$\mu\mathbb{E}^{\mathrm{db}} : \mathbf{K}\,\mu\mathrm{RV}^{\mathrm{db}}[*] \longrightarrow !\mathbf{K}\,\mathrm{RES}[*]$$

such that $(\boldsymbol{P}_\Gamma)$ *vanishes and, for all* $x \in \mathbf{K}\,\mu\mathrm{RES}[*]$, $\mu\mathbb{E}^{\mathrm{db}}(x) = \phi(x)$.

Proof. Let $F : \mathbf{K}\,\mu\mathrm{RV}^{\mathrm{db}}[*] \longrightarrow \mathbf{K}\,\mathrm{RV}[*]$ be the forgetful homomorphism and set

$$\mu\mathbb{E}^{\mathrm{db}} = \widetilde{\mathbb{E}}_b \circ F.$$

This is graded, and its restriction to $\mathbf{K}\,\mu\mathrm{RES}[*]$ is ϕ. On the bounded tensor product its assignment is $\phi \times (\psi^{\mathrm{db}} \circ \lambda)$, by Remark 4.14. Thus it remains to compute the radius generators. If $\gamma = 0$,

the annulus is empty and the value-zero singleton classes $[\{t_0\}]$ and $[1]$ agree, so $\boldsymbol{P}_0 = 0$. For an indexed radius $\gamma > 0$, we have

$$\mu\Psi^{\mathrm{db}}\big(1 \otimes [(0,\gamma)] + 1 \otimes [\{\gamma\}] + [\{t_\gamma\}] \otimes 1 - [1] \otimes 1\big) = \boldsymbol{P}_\gamma.$$

Its image under $\phi \otimes (\psi^{\mathrm{db}} \circ \lambda)$ is

$$\chi((0,\gamma))[\mathbb{G}_m] + \chi(\{\gamma\})[\mathbb{G}_m] + [1] - [1] = -[\mathbb{G}_m] + [\mathbb{G}_m] = 0.$$

The open-interval-plus-endpoint expansion is used only for $\gamma > 0$.

Resolved referee R071 [Error] – The zero-radius case in the proof.
Original source line 1657. The displayed decomposition and its vanishing computation apply only for $\gamma > 0$. At $\gamma = 0$, $\boldsymbol{P}_0 = 0$, but the displayed right side leaves the nonzero term $[\{0\}]$ in the $\Gamma[1]$ factor, whose image is $[\mathbb{G}_m]$. Handle $\gamma = 0$ separately, and for $\gamma > 0$ write the calculation explicitly: $\chi((0,\gamma))[\mathbb{G}_m] + \chi(\{\gamma\})[\mathbb{G}_m] + [1] - [1] = -[\mathbb{G}_m] + [\mathbb{G}_m] = 0$. The artificial convention $o_0 = -p_0$ concerns functions; it does not change the actual class of the empty interval $(0,0)$.

Author resolution. The zero generator is handled directly, and the interval-plus-endpoint calculation is restricted to positive radii. The intended bounded retraction is unchanged.

□

The composition of $\mu\mathbb{E}^{\mathrm{db}}$ with the obvious forgetful homomorphism $!\mathbf{K}\,\mathrm{RES}[*] \longrightarrow !\mathbf{K}\,\mathrm{RES}$ is denoted by $\mathbb{E}^{\diamond}$. By Remark 4.14, the diagram commutes:

$$\begin{array}{ccc} \mathbf{K}\,\mu\mathrm{RV}^{\mathrm{db}}[*] & \longrightarrow & \mathbf{K}\,\mathrm{RV}[*] \\ & {\scriptstyle \mathbb{E}^{\diamond}} \searrow \quad \swarrow {\scriptstyle \mathbb{E}_b} & \\ & !\mathbf{K}\,\mathrm{RES} & \end{array} \tag{4.8}$$

Here the arrow labelled $\mathbb{E}_b$ means the quotient map $\mathbf{K}\,\mathrm{RV}[*] \longrightarrow \mathbf{K}\,\mathrm{RV}[*]/(\boldsymbol{P} - 1)$, followed by (4.3) and the degree-zero point-localized identification in Remark 4.4.

Resolved referee R072 [Clarification; revised] – Identify the map labelled E_b.
Original source line 1665. Optional clarification of the diagram's shorthand: the right arrow is the quotient map $\mathbf{K}\,\mathrm{RV}[*] \longrightarrow \mathbf{K}\,\mathrm{RV}[*]/(\boldsymbol{P} - 1)$, followed by (4.3) and the degree-zero identification $(!\mathbf{K}\,\mathrm{RES}[*][[1]^{-1}])_0 \cong !\mathbf{K}\,\mathrm{RES}$ from Remark 4.4. This is a valid composite. Its point-localized identification is unaffected by the affine-localized injectivity issue noted above.

Author resolution. The valid composite represented by the diagram arrow is now spelled out. The diagram is retained.

This provides an alternative and more direct (but less comprehensive) construction of $\mathbb{E}^{\diamond}$.

Remark 4.16. Changing $!\mathbf{K}\,\mathrm{RES}$ to $!\mathbf{K}\,\mathrm{RES}[[\mathbb{A}]^{-1}]$ in (4.8), we see that $\mathbb{E}_g$ does not commute with $\mathbb{E}^{\diamond}$: by (4.4),

$$\mathbb{E}_g([1]) = [\mathbb{A}]^{-1} \quad \text{whereas} \quad \mathbb{E}^{\diamond}([1]) = \mathbb{E}_b([1]) = 1.$$

This is but an example of a general phenomenon, see Remark 7.28.

We can certainly redefine $\mathbb{E}^{\diamond}$ as the composition of $\mu\mathbb{E}^{\mathrm{db}}$ with the homomorphism $!\mathbf{K}\,\mathrm{RES}[*] \longrightarrow !\mathbf{K}\,\mathrm{RES}[[\mathbb{A}]^{-1}]$ implicit in Remark 4.4 and thereby make it compatible with $\mathbb{E}_g$, but as we have mentioned above, $\mathbb{E}_b$ is ultimately a better choice; see Remark 8.14.